Ben Lonnqvist

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Education

2020- Ph.D. Candidate, Computational Neuroscience
EPFL (Swiss Federal Institute of Technology in Lausanne) | Lausanne, Switzerland
- Advisors: Prof. Michael H. Herzog, Prof. Martin Schrimpf
- Primary interests: NeuroAI; computational neuroscience, psychophysics
- Found key differences in perception between humans and AI
- Core contributor to the open source science project Brain-Score

2016-2020 M.A. Hons, Economics and Finance
University of Aberdeen | Aberdeen, Scotland
- First-class Honours degree; thesis: Optimal search in multi-armed bandit problems
- Relevant coursework: Econometrics, Mathematical and Statistical Methods in Economics, Stochastic processes, Understanding Statistics, Proof-based Microeconomics

Research Output

Publications

2024 **Lonnqvist, B.***, Wu, Z.*, Herzog, M. H. (2024). Latent Noise Segmentation: How Neural Noise Leads to The Emergence of Segmentation and Grouping. *International Conference on Machine Learning (ICML)*. https://proceedings.mlr.press/v235/lonnqvist24a.html.

2021 **Lonnqvist, B.**, Bornet, A., Doerig, A., Herzog, M. H. (2021). A comparative biology approach to DNN modeling of vision: A focus on differences, not similarities. Publisher: *Journal of Vision*. DOI: https://doi.org/10.1167/jov.21.10.17

2020 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Crowding in humans is unlike that in convolutional neural networks. Publisher: *Neural Networks*. DOI: https://doi.org/10.1016/j.neunet.2020.03.021

Pre-prints

2022 **Lonnqvist, B.**, Machiraju, H., Herzog, M. H. (2022). A comment on Guo et al. (2022). arXiv preprint. DOI: https://doi.org/10.48550/arXiv.2208.01456.

2020 **Lonnqvist, B.**, Elsner, M., Hunt, A. R., Clarke, A. D. F. (2020). Modeling individual variation in visual search with reinforcement learning. PsyArXiv. DOI: https://doi.org/10.31234/osf.io/suj28.

Conference oral presentations

2024 **Lonnqvist, B.**, Wu, Z., Herzog, M. H. (2024). Perceptual Grouping with Latent Noise. Oral presentation at the *24th Annual Meeting of the Vision Sciences Society (VSS)*.

2024 **Lonnqvist, B.**, Wu, Z., Herzog, M. H. (2024). Perceptual Grouping with Latent Noise. Oral presentation at the MODVIS Workshop of the *24th Annual Meeting of the Vision Sciences Society (VSS)*.

2023 **Lonnqvist, B.**, Wu, Z., Herzog, M. H. (2023). How object segmentation and perceptual grouping emerge in noisy variational autoencoders. Oral presentation at the *MODVIS Workshop of the 23rd Annual Meeting of the Vision Sciences Society (VSS)*.

2022 **Lonnqvist, B.**, Bornet, A., Doerig, A., Herzog, M. H. (2022). Global Information Processing in Feedforward Deep Networks. Oral presentation at the *22nd Annual Meeting of the Vision Sciences Society (VSS)*.

2019 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Object Recognition in Deep Convolutional Neural Networks is Fundamentally Different to That in Humans. Oral presentation at the *18th annual meeting of the Scottish Vision Group (SVG)*.

Conference poster presentations

2024 **Lonnqvist, B.**, Scialom, E., Merchant, Z., Herzog, M. H., Schrimpf, M. (2024). Current DNNs are Unable to Integrate Visual Information Across Object Discontinuities. Poster presentation at the *7th Annual*

Conference on Cognitive Computational Neuroscience (CCN).

2024 **Lonnqvist, B.***, Wu, Z.*, Herzog, M. H. (2024). Latent Noise Segmentation: How Neural Noise Leads to The Emergence of Segmentation and Grouping. *International Conference on Machine Learning (ICML)*. https://proceedings.mlr.press/v235/lonnqvist24a.html.

2024 Scialom, E.*, **Lonnqvist, B.***, Merchant, Z., Schrimpf, M., Herzog, M. H. (2024). Psychophysics Reveals a Failure of Grouping in Current Deep Neural Networks. Poster presentation at the *46th Annual European Conference on Visual Perception (ECVP)*.

2024 Scialom, E., Merchant, Z., Ernst, U. A., Rotermund, D., **Lonnqvist, B.**, Herzog, M. H. (2024). Categorical object properties outweigh local visual information in object recognition. Poster presentation at the *24th Annual Meeting of the Vision Sciences Society (VSS)*.

2023 **Lonnqvist, B.**, Wu, Z., Herzog, M. H. (2023). How object segmentation and perceptual grouping emerge in noisy variational autoencoders. Poster presentation at the *23rd Annual Meeting of the Vision Sciences Society* (VSS).

2021 **Lonnqvist, B.**, Doerig, A., Bornet, A., Francis, G., Schmittwilken, L., Herzog, M. H. (2021). How crowding challenges (feedforward) convolutional neural networks. Poster presented at the *21st Annual Meeting of the Vision Sciences Society (VSS)*. DOI: https://doi.org/10.1167/jov.21.9.2039.

Invited talks

2024 Scene Grammar Lab, Goethe Universität Frankfurt (Virtual)

2023 INCC, Université Paris Cité & CNRS Vision Group Symposium, Paris, France

2019 Psychology & Computer Science Research Groups, University of Essex, Britain, UK

Research Work Experience

Aug - 2020 - **EPFL (Swiss Federal Institute of Technology in Lausanne) | Doctoral Assistant**

- Doctoral Research Assistant in the NeuroAI Laboratory, and the Laboratory of Psychophysics
- Core contributor to the open-source Python project Brain-Score: https://github.com/brain-score
- Focus on deep learning and psychophysics

Mar - May 2018 - *2020* **Aberdeen School of Psychology | Undergraduate Research Assistant**

- Voluntary research assistant in the Consciousness, Attention, and Perception Lab
- First-authored a paper that has now been published in *Neural Networks*
- Using Python, Keras and NumPy, programmed several architectures of Deep Neural Networks and wrote testing environments to produce novel research on visual crowding in convolutional NNs
- Used SSH tunneling to access a Linux high-power computing cluster

May - Aug 2019 - *2019* **Aberdeen School of Psychology | Research Intern**

- Summer research intern at the Eye Movements and Attention Lab
- Using Python, modeled Bayes-optimal eye movements parameterized by human visual search data
- Using MATLAB and PsychToolbox, wrote a visual search psychophysical experiment
- Ran 65 participant-hours of eye tracking experiments using EyeLink hardware

Selected Awards

2019 Rank Prize Foundation Optoelectronics Studentship

Teaching and Students

2023-2024	Understanding Statistics	Teaching assistant *(MSc course on statistics)*
2021	Real Analysis	Teaching assistant *(BSc course on mathematics)*
2020	Real Analysis	Teaching assistant

2024	Emily Nurden	EPFL Summer Research Program
2023-2024	Maya van Holk	MSc thesis *(co-advised with Elsa Scialom)*
2023-2024	Maisa Ben Salah	MSc thesis *(co-advised with Martin Schrimpf)*
2023	Clémentine Lévy-Fidel	MSc thesis
2023	Ismaïl Sahbane	ML4Science project *(MSc CS course project)*
2023	Jad Tala	ML4Science project
2023	Olena Zavertiaiva	ML4Science project
2023	Pauline Verchinine	ML4Science project
2023	Jennifer Ayer	ML4Science project
2023	Laurent Brock	ML4Science project
2022-2023	Zhengqing Wu	MSc Computational Neuroscience Minor Thesis
2022-2023	Merkourios Simos	MSc Semester Project

Academic service

Reviewing

Journals *Neural Networks, Neural Computation, PLOS One, Vision Research*

Conferences *ICML Workshop on LLMs and Cognition, NeurIPS Workshop on Behavioral ML, Cognitive Computational Neuroscience*

Open Source and Open Science Contributions

I am a proponent of fully transparent open science and data, as well as open-source software. I am a key contributor of the Open Source, Open Science NeuroAI modeling initiative Brain-Score, which has $500+$ registered users from almost 200 institutions, and over $130,000+$ website visits in 2024.

I have added several human data benchmarks, and made core contributions and improvements to the library, leading $10+$ merged pull requests and contributing $300+$ commits.